



Ophiopogonin A alleviates acute lung inflammation via NF- κ B/MAPK signaling pathway in a mouse model of *Klebsiella pneumoniae*

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ABSTRACT

Klebsiella pneumoniae (Kp) is a gram-negative bacterium present in the flora of the mouth, skin and intestines. Due to the emergence of antibiotic resistance, the outcome of Kp pneumonia is becoming worse. Exploring an effective treatment method is clinically important. *Ophiopogon japonicus* is a traditional Chinese herbal medicine for treating inflammation and oxidative stress damage, as ophiopogonin A (OP-A) is its main active ingredient. We investigated the impact of OP-A on Kp pneumonia and further elucidated the underlying mechanism. After the establishment of Kp mouse model, the animals were grouped and received intraperitoneal injection of OP-A, levofloxacin (LvX) or PBS. After 12 hours, mice were dissected to detect the pathological changes of lung tissues, and the number of inflammatory proteins and inflammatory cells in the bronchoalveolar lavage fluid (BALF). Finally, Western blot and RT-qPCR analyses were carried out to detect MARK, JNK, ERK and NF- κ B expression in mouse lung tissues upon treatments. Administration of OP-A attenuated the pathological damage of lung tissues of Kp pneumonia mice, as both LvX and OP-A significantly controlled and inhibited the wet-to-dry ratio of the lung tissues. Increased protein content and inflammatory cells infiltration were visible in Kp pneumonia mice, while the advent of LvX and OP-A dramatically diminished inflammatory cells infiltration in BALF of lung tissues with fewer contents of IL-1 β , IL-6 and TNF- α . The protective effect of OP-A on Kp pneumonia correlated with the NF- κ B/MAPK signaling. In conclusion, OP-A treatment may alleviate the pathological damage and reduce inflammation of Kp mice through inhibition of the NF- κ B/MAPK signaling pathway. OP-A has a significant anti-inflammatory effect on Kp mice, and hence it is a promising alternative for treating Kp pneumonia.

Keywords: Ophiopogonin A, NF- κ B/MAPK, *Klebsiella pneumoniae*, Acute Pneumonia.

1. INTRODUCTION

Klebsiella pneumoniae (Kp) is a gram-negative bacterium present in the flora of the mouth, skin, and intestines and it is also the third most prevalent microorganism identified in blood cultures of patients with sepsis [1]. Accompanied with antibiotic resistance, Kp infection is becoming a major threat to human health [2]. Pneumonia is the most common infection caused by Kp, usually manifested as bronchopneumonia and bronchitis with a high mortality

(about 50%) even under antibacterial treatment [3]. However, despite advances in etiology, the underlying pathogenesis of Kp infection and host's molecular mechanism of defense in the lungs are still elusive. It is believed that Kp infection is the result of synergistic effect of many factors.

NF- κ B/MAPK signaling regulates inflammation of mice with Kp, and inhibition of this may decrease inflammatory factors release and suppress neutrophils recruitment [4]. Activation of this pathway may promote cell inflammation and increase the severity of Kp infection [5]. NF- κ B, a transcriptional regulator, is known to contribute to

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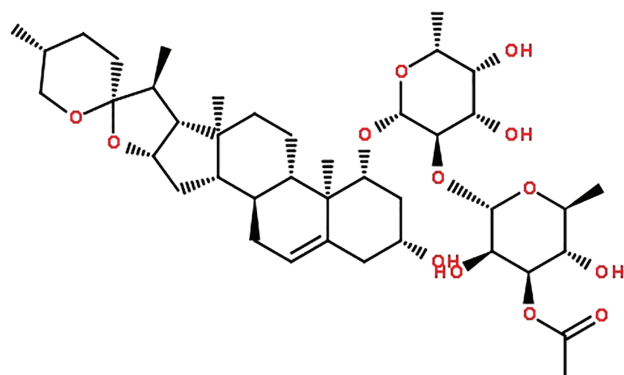


Fig. 1. Molecular formula of Ophiopogon saponin A.

the regulation of chronic inflammation in the body. NF- κ B family includes p50, p52, p65 and RelB, among which p65-p50 heterodimer are most active form [6, 7]. Activation of IKK induces degradation of proteasome I κ B, accompanied with production of NF- κ B subunits to initiate inflammation-related gene expression [8]. The MAPK pathway makes a difference to the inflammation process. Collectively, the phosphorylation of ERK, JNK and p38 may induce the expression of some inflammatory factors [9–11]. Therefore, inhibition of NF- κ B/MAPKs signaling may be a way for treating inflammatory diseases.

Investigation on various medicines is always a hot topic [12–18]. *Ophiopogon japonicus* is a traditional Chinese herbal medicine for inflammation and oxidative stress damage. Accumulating evidence has depicted the significant anti-inflammatory and antioxidant effects of *Ophiopogon japonicus*. Besides, it reduces the aggregation of macrophages and neutrophils, down-regulates the expression of pro-inflammatory cytokines, and protecting endothelial cells from apoptosis and promoting microcirculation [19–21]. Ophiopogon A (OP-A) is the main ingredient of *Ophiopogon japonicus* with anti-inflammatory and antioxidant effects [22]. OP-A is a steroidal glycoside compound derived from *Ophiopogon japonicus* (Fig. 1). Evidence has noted the impact of OP-A on biological properties, including venous thrombosis with anti-inflammatory and antitussive activities [23]. In this study, using a mouse model of Kp pneumonia, we explored the protective mechanism of OP-A in Kp pneumonia. Our results proved that OP-A restrained the development of Kp pneumonia through the NF- κ B/MAPK signaling. The findings might promote the understanding on the function of OP-A, and help development of treatment for acute inflammation.

2. MATERIALS AND METHODS

2.1. Reagents

OP-A was from Sigma and Levofloxacin (LvX) from Chengdu Kelun Co., Ltd. The MPO detection kit

was obtained from Beyotime Biotechnology Co., Ltd. (Shanghai, China). The ELISA kit was provided by Thermo Fisher. Antibodies to JNK, p-21 ERK, ERK, p-P38, P-38, β -Tublin, and p-JNK were acquired from Santa Cruz Biotechnology (Boston, MA, USA), as well as HRP-labeled anti-rabbit or anti-mouse IgG.

2.2. Animals

Forty adult female BALB/c mice (18–20 g) (Shanghai Laboratory Animal Center, Shanghai, China) were given food and water normally and placed in a suitable living environment. All experiments followed the International Guidelines for Biomedical Research Involving Animals and was approved by the Animal Use Committee of Wuhan University.

2.3. Model Establishment

Kp acquired from Wuhan Institute of Microbiology was cultured on agarose gel medium plates. Kp was diluted with PBS for animal model induction. All mice were assigned into control group, model group, OP-A treatment group, and LvX treatment group. Under anesthesia with 10% chloral hydrate, the mice of control group were injected with PBS 0.1 mL via tracheal intubation; the other mice were injected. One hour later, the OP-A treatment group and the LvX treatment group were injected intraperitoneally with OP-A (80 mg/kg) and LvX (31.64 mg/kg), respectively. After 12 hours, all animals were sacrificed for following experiments.

2.4. H&E Staining

The left lower lobe of the left lung was fixed in 4% pre-chilled paraformaldehyde, dehydrated with gradients of ethanol, embedded in paraffin and then stained with H&E as previously described [10]. The pathological changes of all groups were observed.

2.5. Wet/Dry Weight Ratio of Lung Tissues

Lung tissues were removed 12 hours after treatments and wet weight were measured. After lungs were placed at 60 °C for 24 hours, the samples were weighed and the wet-to-dry ratio was calculated.

2.6. Western Blot

Lung tissue was lysed and centrifuged at 12000–16000 rpm for 20 mins. With concentration detected by BCA method, proteins were separated by 10% SDS-PAGE electrophoresis and transferred to a PVD, and then blocked with 5% skimmed milk. The membrane was probed with primary antibodies against p-JNK, JNK, p-21 ERK, ERK, p-P38, P-38, β -Tublin, GAPDH 24 (Santa Cruz) overnight and then with secondary antibodies. The blot was developed by a chemiluminescence (ECL) kit.

2.7. RT-qPCR

RNA was extracted and the agarose gel electrophoresis and absorbance measurement were performed with a Thermo BIONANO ultra-micro ultraviolet spectrophotometer (Thermo Forma, MA, USA) at the wavelengths of 260 and 280 nm. cDNA was synthesized and amplified with Quantifast SYBR Green PCR kit and the reaction system. NF- κ B and GAPDH primers were acquired from Shanghai Generay Biotechnology Co., Ltd.

NF- κ B: F(5'-3'): ACGATCTGTTTCCCCTCATCT,
R(5'-3'): TGGGTGCGTCTT AGTGGTATC;
GAPDH: F(5'-3'): GGTGAAGGTCGGTGTGAACG
R(5'-3'): CTCGCTCCTGGA AGATGGTG

2.8. ELISA

Bronchoalveolar lavage fluid (BALF) was injected with 0.5 mL of PBS. The mixture was centrifuged to collect supernatant. ELISA was conducted to detect the contents of IL-1 β , IL-6, and TNF- α .

2.9. Statistical Method

Graphpad Prism software processed data which were displayed as mean $\pm$ SEM and analyzed by one-way ANOVA. $P < 0.05$ indicates a difference.

3. RESULTS AND DISCUSSION

3.1. Molecular Formula of Ophiopogonin A and Its Effect on Lung Injury Induced by Kp

Ophiopogonin A was extracted from the dry tuber root of Ophiopogon liliaceae (Fig. 1). The pharmacological

effects display as the functions of nourishing lung and promoting fluid, nourishing Yin and clearing heat. To assess OP-A's effect on Kp, we obtained the lower lobe of the left lung for histological examination after 12 hours of treatment (Fig. 2). After injection of Kp, inflammatory cell infiltration and alveolar hemorrhage appeared while the morphology of the lung tissue of the control group remained normal. Treatment with either OP-A or Lvx significantly reduced the above-mentioned pathological changes induced by Kp and alleviated lung injury. Collectively, the above results indicated that OP-A reduced acute lung injury caused by Kp.

3.2. OP-A Reduces Pulmonary Edema During Inflammation and Restores Capillary Permeability

To assess the impact of the drugs on pulmonary edema, we measured wet/dry weight ratio of lung tissues and found it was significantly increased in Kp group. Injection of either OP-A or Lvx decreased wet/dry weight ratio (Fig. 3(A)). The change of total protein concentration in BALF is a characteristic of increased capillary permeability. The protein content of BALF in the Kp group were significantly increased but this increase was reversed by OP-A treatment (Fig. 3(B)).

3.3. OP-A Reduces the Infiltration of Inflammatory Cells in BALF

To further verify the effect of OP-A on pneumonia, inflammatory cell number in BALF was counted and found increased the number of macrophages and neutrophils in

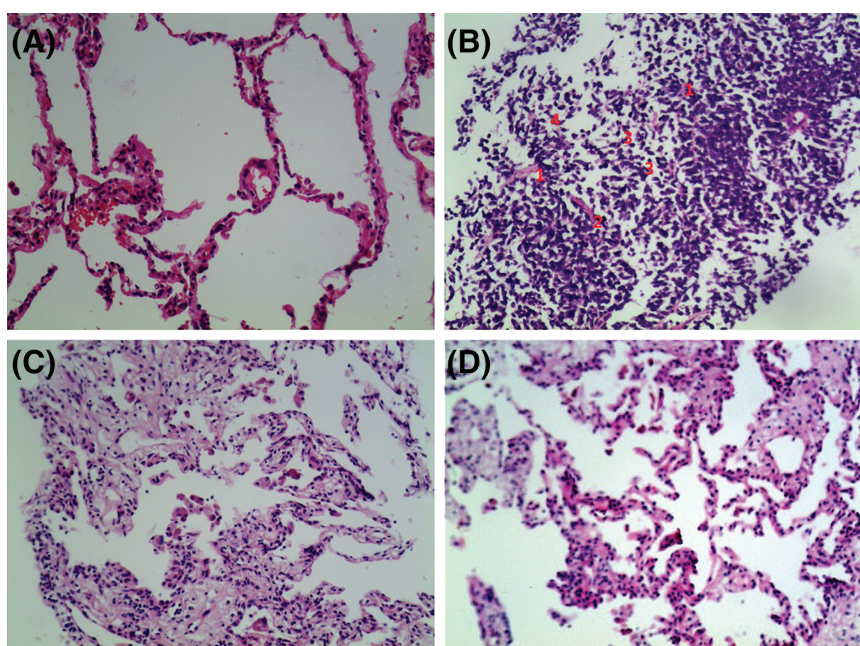


Fig. 2. OP-A mitigates the lung injury induced by Kp ($\times 400$). (A) Control group. (B) Model group. (C) OP-A treatment group. (D) Lvx treatment group. 1: Alveolar congestion 2: Hemorrhage 3: Neutrophil infiltration or aggregation 4: Alveolar wall thickness.

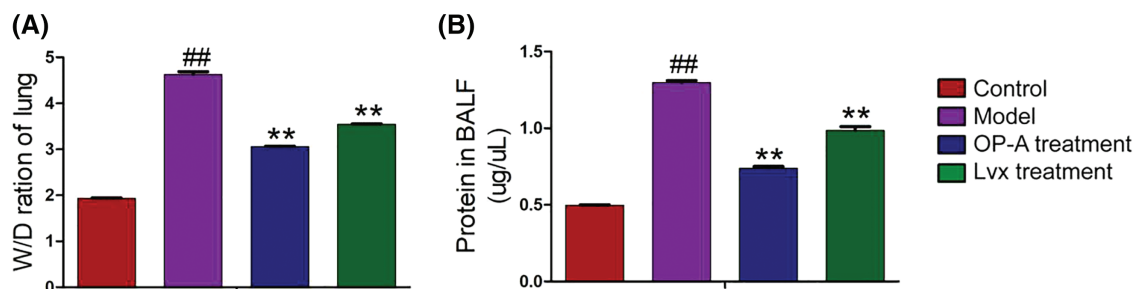


Fig. 3. OP-A mitigates pulmonary edema and restores capillary permeability. (A) The effect of OP-A on the lung wet-to-dry weight ratio. (B) Quantification of protein content in BALF induced by Kp. ^{##} $p < 0.05$ versus the control group. ^{**} $p < 0.01$ versus the model group.

BALF after Kp injection, which was decreased significantly upon treatment with OP-A or Lvx (Fig. 4).

3.4. OP-A Reduces Pro-Inflammatory Cytokines in BALF

As seen in Figure 5, mice of model group had increased level of TNF- α , IL-1 β and IL-6. Of note, OP-A exhibited the same effect as Lvx, which demonstrated as down-regulating pro-inflammatory cytokines and improving tissue damage. These results indicated that OP-A might improve lung injury by reducing the secretion of pro-inflammatory cytokines induced by Kp.

3.5. Effect of OP-A on NF- κ B mRNA

Using RT-qPCR, we found the up-regulation of NF- κ B in model group (Fig. 6). The results indicated that OP-A and Lvx decreased NF- κ B mRNA level.

3.6. The Effect of OP-A on MAPK/JNK/ERK Pathway

To further investigate the underlying mechanism of OP-A in Kp pneumonia, we checked up the MARK/JRN/ERK pathway in the lung tissues of Kp rats (Fig. 7). Using Western Blot analysis, we noticed the up-regulation of p-JNK and p-ERK levels in the model group, but their levels were dramatically reduced by OP-A treatment.

3.7. Discussion

Kp pneumonia is a bacterial respiratory infection and is usually featured with high mortality rate due to its resistance to various antibiotics [24]. After Kp infection, numerous inflammatory cells (such as neutrophils and macrophages) are recruited into infection site in response to injury and infection [25, 26]. However, these inflammatory cells release too much inflammatory factors, which

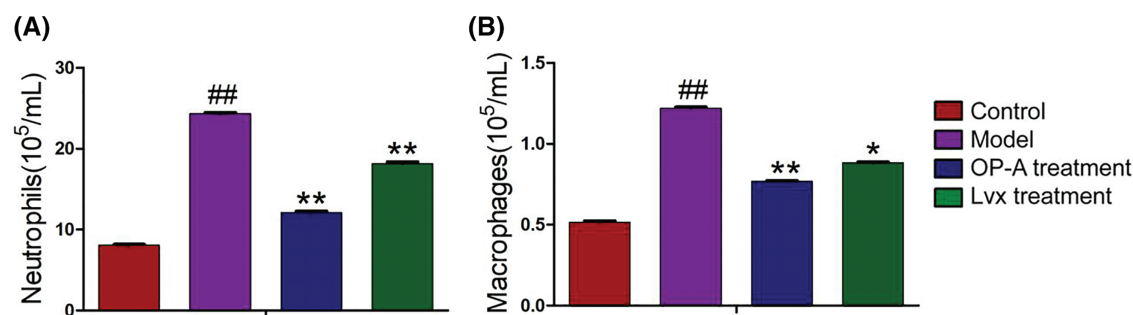


Fig. 4. OP-A reduces the infiltration of inflammatory cells in BALF. (A) Infiltration of neutrophils; (B) Infiltration of macrophages; ^{##} $p < 0.01$ versus the control group. ^{**} $p < 0.01$ and ^{*} $p < 0.05$ versus the model group.

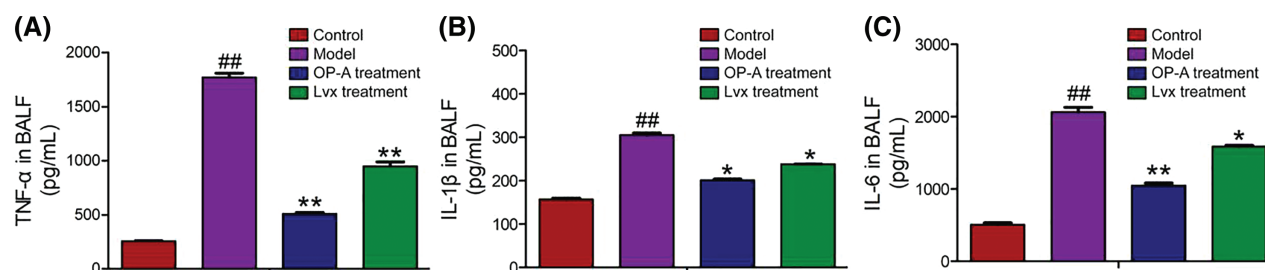


Fig. 5. The effect of OP-A on the secretion of pro-inflammatory cytokines. (A–C) ELISA of TNF- α , IL-1 β and IL-6 contents in BALF. ^{##} $p < 0.01$ versus the control group. ^{**} $p < 0.01$ and ^{*} $p < 0.05$ versus the model group.

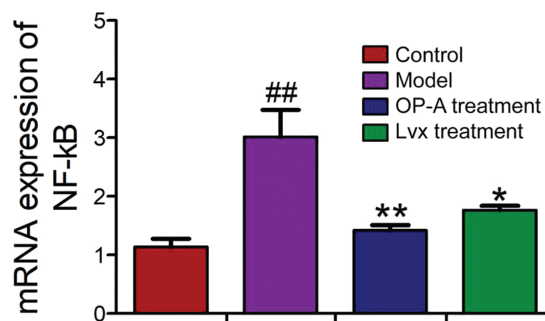


Fig. 6. Effect of OP-A on NF- κ B mRNA expression. ## $p < 0.01$ versus the control group. ** $p < 0.01$ and * $p < 0.05$ versus the model group.

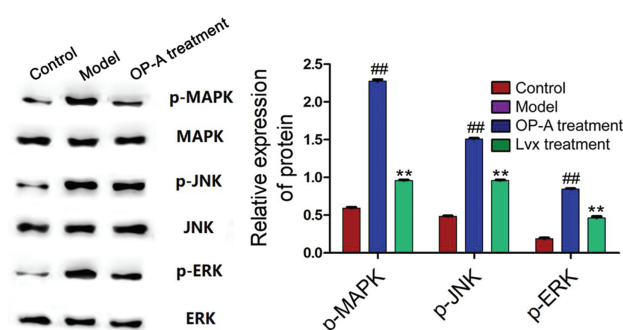


Fig. 7. The effect of OP-A on the MAPK/JNK/ERK pathway. ## $p < 0.05$ and versus the control group. ** $p < 0.01$ versus the model group.

further aggravates inflammation [27, 28]. OP-A has strong anti-inflammatory and antiviral activities [29–31]. This study demonstrated that administration of OP-A improved the pathological injuries on the lung tissues of Kp mice, and stabilized the wet-to-dry ratio of the lung tissues. In addition, in OP-A treatment group, the infiltration of inflammatory cells in the alveolar lavage fluid was reduced, accompanied with decreased levels of inflammatory proteins. Furthermore, the protective effect of OP-A on Kp pneumonia was closely related to NF- κ B/MAPK signaling. Therefore, this study proves that OP-A may effectively improve Kp pneumonia through NF- κ B/MAPK signaling to inhibit inflammation process.

Lung inflammation is one of the main complications of Kp pneumonia [32]. NF- κ B regulates inflammation and its activation by a variety of mediators can combine with pro-inflammatory genes, thereby causing changes in gene expression and amplification of inflammatory responses [33, 34]. The results of our work depicted that OP-A treatment reduced NF- κ B expression in Kp mice.

MAPKs family mainly consists of ERK, JNK and p38MAPK [35, 36]. Phosphorylation of ERK, JNK and p38 can induce the expression of some inflammatory factors [37–39]. Therefore, inhibition of MAPK can reduce pro-inflammatory protein levels. In this study, after injection of Kp into mice, the content of inflammatory cytokines was increased, and the tissue structure was severely damaged. However, OP-A treatment suppressed

the MAPK signaling pathway and decreased the release of inflammatory cytokines.

4. CONCLUSION

In conclusion, OP-A treatment mitigated the lung injuries, inhibited inflammation, and stabilized cell permeability through inhibition of NF- κ B/MAPK signaling. The evidence suggests that with significant anti-inflammatory effect, OP-A may be used as an effective drug for treating Kp pneumonia.

Ethical Compliance

This study was approved by the ethnic committee of Suzhou Hospital of Integrated Traditional Chinese and Western Medicine.

Conflicts of Interest

There are no conflicts to declare.

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